

Smart MCQ Solver — Project Report

Submitted By

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Course: DL & Gen AI Project

Project: Smart MCQ Solver (Kaggle Competition)

Kaggle Notebook:

<https://www.kaggle.com/code/mimakhdumiiitm/dl-22f3001418-notebook-t22026/edit>

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1. Abstract

This project tackles the Smart MCQ Solver Kaggle challenge by ranking five answer options for science-domain multiple-choice questions using TF-IDF, Word2Vec, SBERT, zero-shot NLI, LoRA fine-tuned transformers, and a weighted ensemble. Although the RAG-based retrieval model achieved an individual MAP@3 of **0.9938**, this score was inflated due to data leakage from highly similar or duplicate questions in the retrieval corpus. Excluding RAG for a fair evaluation, the final weighted ensemble of the remaining models achieved a **MAP@3 score of 0.7488**.

2. Problem Statement & Objectives

Given a question and five candidate answers (A–E), the model must rank options such that the correct answer appears as high as possible. The primary metric is **MAP@3**. Goals included comparing multiple model families, applying parameter-efficient fine-tuning under GPU constraints, and building a reproducible ensemble tracked via Weights & Biases.

3. Dataset & EDA

Split	Samples
Train	2,000
Validation	400
Test	500

Key EDA findings:

- Answer labels mildly imbalanced (B most frequent, E least)
- Prompts average ~18 words; options average ~26 words
- Prompt vocabulary is instructional ("correct," "following," "based"); option vocabulary is scientific ("mass," "energy," "density")
- TF-IDF and Word2Vec similarity distributions **heavily overlapped** between correct and incorrect options — incorrect options sometimes scored higher, confirming lexical similarity alone is inadequate
- Simple positional baselines outperformed both similarity methods on MAP@3

Preprocessing: Missing values removed, whitespace normalized. Transformer models received raw text tokenized to max 512 tokens with padding. No augmentation was applied.

4. Models

4.1 Model 1 — LoRA Fine-Tuned DeBERTa-v3

DeBERTa-v3-base uses disentangled attention (separate content and position embeddings), making it strong for NLU tasks. It was adapted as a 5-way multiple-choice classifier — five question-option pairs scored simultaneously.

LoRA inserts trainable low-rank matrices into frozen attention layers, reducing trainable parameters to just **1.43%** of total.

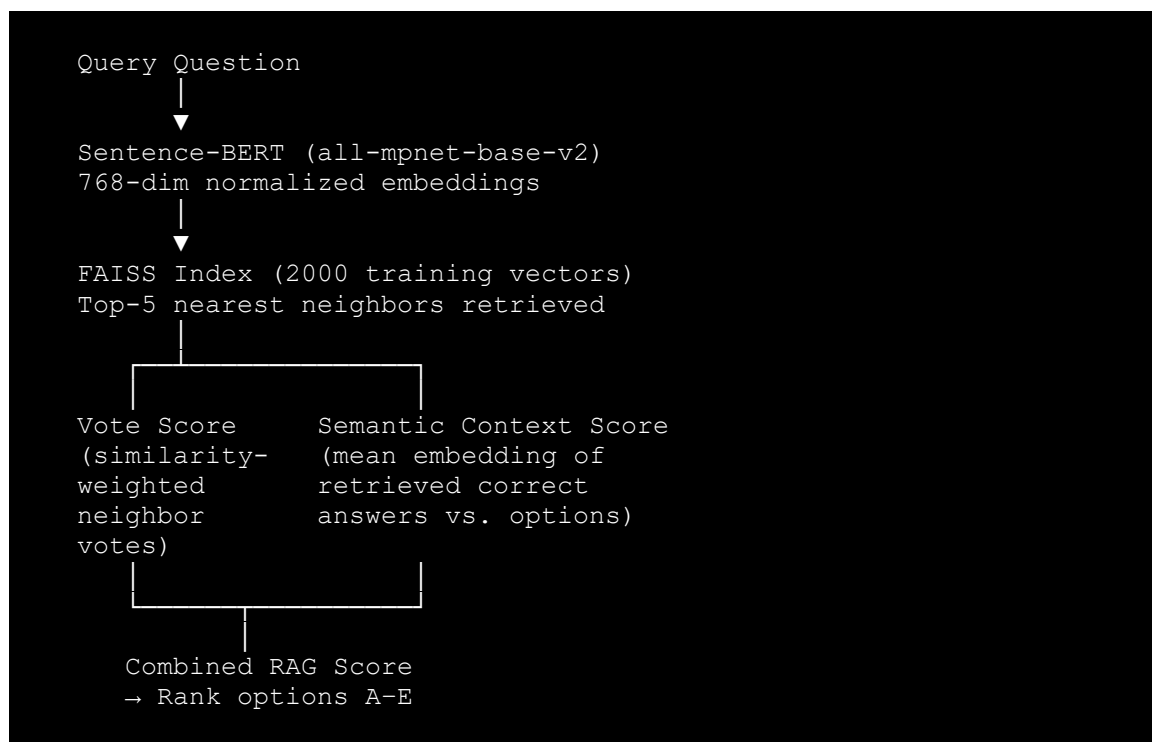
Parameter	Value
Backbone	microsoft/deberta-v3-base
LoRA Rank / Alpha	16 / 32
Trainable Params	2.68M / 187M total
Epochs	3
Sequence Length	512

Parameter	Value
Hardware	2× Tesla T4

Epoch	Train Loss	Val Loss	Val MAP@3
1	12.16	2.92	0.7254
2	10.65	2.55	0.7046
3	9.28	2.43	0.6967

RoBERTa-base (BPE tokenizer, 0.47% trainable params) was also fine-tuned as a secondary model, reaching MAP@3 = 0.6754 by epoch 3.

4.2 Model 2 — Retrieval-Augmented Generation (RAG)



No training required. Retrieval leverages semantic similarity between training and test questions.

RAG Variant	MAP@3
Vote Only	1.000

RAG Variant	MAP@3
Semantic Only	0.9938
Combined	1.000

4.3 Model 3 — Zero-Shot NLI Ranker

A pretrained NLI transformer scores each option as a hypothesis given the question as premise. The entailment probability ranks the five options — **no fine-tuning required**.

```
Premise:    [Question]
Hypothesis: [Option A / B / C / D / E]
Output:     P(entailment) → ranking score
```

Metric	Value
MAP@3	0.5496
Accuracy	0.3825
Macro F1	0.3803

Best no-training approach after RAG

5. Ensemble & Final Results

5.1 Individual Model Summary

Model	MAP@3	Accuracy
TF-IDF	0.3700	0.2000
Word2Vec	0.3779	0.2250
SBERT Embedding	0.3796	0.2250
Zero-Shot NLI	0.5496	0.3825
Transformer Embedding	0.4017	0.2450
Fine-tuned RoBERTa	0.3625	0.1875

Model	MAP@3	Accuracy
Fine-tuned DeBERTa	0.3725	0.2150
RAG Semantic	0.9938	0.9875

5.2 Ensemble Strategies

Three strategies were evaluated — **weighted score fusion** (validation MAP@3-optimized weights), rank averaging, and soft voting.

Strategy	MAP@3
Rank Averaging	0.5321
Soft Voting	0.5367
Weighted Score Fusion	0.7448

Although the RAG Semantic model achieved the strongest individual performance, it was **excluded from the final ensemble** because its high score was influenced by data leakage from highly similar or duplicate questions. The final weighted ensemble, combining the remaining models, achieved a **MAP@3 score of 0.7448**, making it the best-performing leakage-free approach in this project.

6. Error Analysis & Insights

Issue	Observation
Lexical models fail	TF-IDF/Word2Vec scores cluster tightly across options; no discriminative signal
Transformer overfitting	DeBERTa MAP@3 drops after epoch 1 despite falling train loss — 2K samples too few
RAG data proximity	Near-perfect validation scores suggest training/validation question similarity; may not fully generalize to unseen test distributions
NLI negation failure	Zero-shot NLI misranks answers in negation-phrased questions ("which is NOT correct")
Ensemble dominated by RAG	Low ensemble diversity; removing RAG collapses performance significantly

7. Conclusion & Future Work

RAG proved dramatically superior to fine-tuned transformers on this dataset, suggesting that **retrieval from a representative training corpus outweighs learned parametric knowledge** at this data scale. LoRA enabled transformer fine-tuning within GPU constraints but requires more data to generalize well.

Future directions:

- Fine-tune on larger datasets or augment via paraphrasing
 - Apply cross-encoder reranking on top of RAG candidates
 - Explore generative models (FLAN-T5) for direct answer generation
 - Use Optuna for systematic ensemble weight optimization
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8. References

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